

1. Title of the Project

Raphaaa Ecommerce Website: Multi-Role Full-Stack Commerce and Operations Platform

2. Internship Details

- Internship Domain: Full Stack Web Development (Ecommerce + Operations)
- Project Type: Industry-oriented B2C Ecommerce Platform
- Technology Nature: Web-based, modular, role-driven, scalable

3. Introduction

Modern ecommerce businesses require more than a storefront. They need integrated systems for product management, order processing, payments, customer engagement, marketing campaigns, and internal team operations. In many organizations, these functions are distributed across separate tools, causing inefficiency and data fragmentation.

The Raphaaa Ecommerce Website was developed to solve this by combining customer commerce workflows and internal business operations in one role-based platform. The project supports customer shopping journeys and also provides dedicated operational interfaces for admin, merchantise, delivery, and marketing teams.

4. Problem Statement

Common problems in conventional ecommerce setups include:

- Disconnected product, order, and marketing systems.
- Limited role-based control for internal teams.
- Poor visibility into inventory, sales trends, and revenue.
- Delayed communication for campaigns, offers, and customer updates.
- Lack of centralized handling for support requests, complaints, and tasks.

Therefore, a unified and scalable ecommerce platform was required to improve customer experience, streamline operations, and enhance business decision-making.

5. Objectives

The key objectives of the internship project were:

- To design and implement a full-stack ecommerce platform with role-based access.
- To develop complete customer commerce flow: catalog, cart, checkout, payment, and order tracking.
- To provide admin and operational modules for products, users, orders, inventory, and analytics.
- To integrate secure payment, media storage, and communication services.
- To implement campaign, offer, and subscriber management for marketing workflows.
- To support internal process efficiency using task and complaint modules.

6. Scope of Work

During the internship, work was carried out across:

- Frontend user and admin panel development.
- Backend API design and route-level module implementation.
- MongoDB schema design and role-aware data handling.
- Integration with Razorpay, Cloudinary, email, and OTP services.
- Development of analytics/operations features (inventory, revenue, trends, tasks).
- Functional testing of major role-based workflows and module interaction.

7. Methodology

The project followed a practical module-first approach:

- Requirement analysis for customer and internal user roles.
- Backend route and model structuring for independent module growth.
- Frontend integration with protected routes and role guards.
- Iterative testing and debugging for payment, order, and admin operations.
- Environment-based configuration for deployment readiness.

8. Technology Stack

- Frontend: React, Vite, Redux Toolkit, React Router, TailwindCSS

- Database: MongoDB
- Integrations: Razorpay, Cloudinary, Email services, SMS OTP utilities
- Analytics/UI: Chart.js / Recharts

9. Major Functional Modules Implemented

- Authentication, authorization, and profile management
- Product catalog and collection management
- Cart, checkout, and order management
- Payment integration and webhook processing
- Wishlist and review modules
- Offers and campaign tracking
- Website CMS settings (hero, about, contact, privacy policy)
- User, product, and admin order management
- Inventory, sales trend, and revenue analysis
- Task management and scheduled status automation
- Subscriber, contact, and complaint handling

10. Role-wise Functional Coverage

10.1 Customer

- Browse products and collections
- Add to cart/wishlist and place orders
- Track own orders and submit product reviews
- Manage profile and addresses

10.2 Admin

- Full control of users, products, orders, inventory, and analytics
- Website content and operational settings management
- Oversight for offers, complaints, and internal workflows

10.3 Merchantise

- Product and order operations
- Inventory and business dashboards
- Support for offers, tasks, and operational updates

10.4 Delivery Boy

- Operational order visibility/actions (as role-permitted)
- Profile management and assigned workflow participation

10.5 Marketing

- Campaign creation/tracking and broadcast workflows
- Subscriber and contact lead handling
- Promotion monitoring and content-aligned operations

11. Key Outcomes

- Delivered a production-oriented ecommerce platform with multi-role governance.
- Unified storefront and business operations under one architecture.
- Enabled data-driven decision support through sales and revenue modules.
- Improved campaign and communication workflows via dedicated marketing tooling.
- Built a maintainable, extensible full-stack codebase.

12. Internship Learning Outcomes

The internship provided strong practical exposure to:

- Full-stack architecture and module orchestration
- Role-based security and access control design
- Payment and cloud service integration
- Building operational dashboards and analytics-backed features
- Handling real-world workflow complexity in ecommerce systems

13. Future Scope

- Stronger audit logs and fine-grained permission controls
- Improved automation for delivery lifecycle events
- Advanced recommendation and personalization engine
- Expanded notification and campaign intelligence features

14. Conclusion

The internship project successfully implemented Raphaana Ecommerce Website, a comprehensive full-stack platform that combines customer purchasing journeys with business operations and marketing execution. The solution demonstrates practical industry relevance, scalability potential, and strong alignment with modern ecommerce requirements.